Definitions

Definition 1.5: π-systems

A class $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ is called a *π-system* if it is closed under finite intersections.

Definition 1.6: λ-systems

A class $\mathcal{D} \subseteq \mathcal{P}(\Omega)$ is called a *λ-system* if it contains Ω and is closed under proper differences and increasing limits. That is, if $A, B \in \mathcal{D}$ with $B \subseteq A$, then $A \backslash B \in \mathcal{D}$, and if $A_1, A_2, \ldots \in \mathcal{D}$ with $A_1 \uparrow A$ (that is; $A_1 \subseteq A_2 \subseteq \cdots \subseteq A = \cup_i A_i$), then $A \in \mathcal{D}$.

Definition 2.4: Cartesian Product

Suppose for some index set T, we have sets X_t, $t \in T$. If $\sqcup_{t\in T} X_t$ is the disjoint union of these sets, then the *Cartesian product*, $\bigtimes_{t\in T} X_t$ is the set of maps f from T to $\sqcup_{t\in T} X_t$ such that $f(t) \in X_t$ for all $t \in T$.

Definition 2.5: Product Topology / Tychonoff Topology

Given topological spaces S_t, $t \in T$ for some index set T, the *product topology* (or *Tychonoff topology*) on $\bigtimes_{t\in T} S_t$ is such that open sets in $\bigtimes_{t\in T} S_t$ can be written as the union of sets of the form $\bigtimes_{t\in T} \mathcal{U}_t$ where $\mathcal{U}_t$ are open in S_t and $\mathcal{U}_t = S_t$ for all but finitely many $t \in T$.

In other words, the product topology are generated by sets of the form $\bigtimes_{t\in T} \mathcal{U}_t$ with the properties mentioned.

Definition 2.7: Product σ-Field

Let Ω_t, $t \in T$ be measurable spaces, each with a σ-field $\mathcal{A}_t$. Then, for the Cartesian product, $\bigtimes_{t\in T} \Omega_t$, the *product σ-field* on $\bigtimes_{t\in T} \Omega_t$, denoted $\bigotimes_{t\in T} \mathcal{A}_t$ is the σ-field generated by the *cylinder sets* $A_t \times \bigtimes_{s\neq t} \Sigma_s$ for $A_t \in \mathcal{A}_t$.

Definition 4.2: Measurability of Functions

Given two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a map $f : S \to T$ is said to be *$\mathcal{S}\backslash\mathcal{T}$-measurable* if $f^{-1}(\mathcal{T}) \subseteq \mathcal{S}$

Definition 6.1: f-measurable

Let $f : \Omega \to X$ and $g : \Omega \to Y$ be maps. Then, we say g is *f-measurable* if g is $\sigma(f)$-measurable.

Definition 6.2: Borel Isomorphims and Borel Spaces

Two measurable spaces S and T are *Borel isomorphic* if there exists a bijection $f : S \to T$ such that f and f^{-1} are measurable. A measurable space that is Borel isomorphic to a Borel subset of $[0, 1]$ is called a *Borel space*.

Definition 6.3: Polish Spaces

A *Polish space* is a topological space that has a separable and complete metrization.

Definition 7.3: Push-Forward Measures

Let $f : S \to T$ be measurable on two spaces $(S, \mathcal{S}), (T, \mathcal{T})$. Given any measure μ on $(S, \mathcal{S})$, we can define a measure $\mu \circ f^{-1}$ on $(T, \mathcal{T})$ by

$$(\mu \circ f^{-1})(B) = \mu(f^{-1}(B)) = \mu\{s \in S : f(s) \in B\}$$

for $B \in \mathcal{T}$. We call $\mu \circ f^{-1}$ the *push-forward measure of μ by f*.

Definition 8.5: μ-completions

Given a measure space $(\Omega, \mathcal{A}, \mu)$ and a σ-field $\mathcal{F} \subseteq \mathcal{A}$, define the *$\mu$-completion* of $\mathcal{F}$ in $\mathcal{A}$ as the σ-field $\mathcal{F}^\mu = \sigma(\mathcal{F}, \mathcal{N}_\mu)$, where $\mathcal{N}_\mu$ denotes the class of all subsets of arbitrary μ-null sets in $\mathcal{A}$.

Definition 9.2: σ-finiteness

A measure μ on a measurable space $(\Omega, \mathcal{A})$ is *σ-finite* if $\Omega = \bigcup_{i=1}^{\infty} A_n$ with $A_i \cap A_j = \emptyset$ for $i \neq j$ such that $\mu A_n < \infty$ for all n. Equivalently, μ is σ-finite if, and only if, there is a measurable strictly positive function f such that $\mu f < \infty$.

Definition 9.5: L^p Spaces

Given a measure space $(\Omega, \mathcal{A}, \mu)$ and a $p > 0$, write $L^p = L^p(\Omega, \mathcal{A}, \mu)$ for the class of all measurable functions $f : \Omega \to \mathbb{R}$ with

$$\|f\|_p := \left[\int |f|^p \, d\mu\right]^{1/p} < \infty$$

Definition 12.1: Bounded Signed Measures

A *bounded signed measure* on a measurable space $(\Omega, \mathcal{A})$ is a bounded function $\nu : \mathcal{A} \to \mathbb{R}$ such that

$$\nu\left(\bigcup_n A_n\right) = \sum_n \nu A_n$$

for disjoint $A_i \in \mathcal{A}$, where this series converges absolutely.

Definition 12.2: Mutually Singular Measures

We say μ and ν on $(\Omega, \mathcal{A})$ are *mutually singular* or *orthogonal*, and write $\mu \perp \nu$, if there exists $A \in \mathcal{A}$ with $\mu A = \nu A^c = 0$.

Definition 12.6: Absolute Continuity

For any two measures μ and ν on $(\Omega, \mathcal{A})$, ν is said to be *absolutely continuous with respect to* μ if $\mu A = 0$ implies $\nu A = 0$ for all $A \in \mathcal{A}$, in which case we write $\nu \ll \mu$.

Definition 13.2: Probability Spaces

Fix a probability space $(\Omega, \mathcal{A}, P)$, where P is a *probability measure*; that is, a measure with total mass 1. We call the sets $A \in \mathcal{A}$ *events*, and $PA = P(A)$ is the *probability* of A.

Definition 13.3: Infinitely Often and Ultimately

Given a sequence of events $A_1, A_2, \ldots \in \mathcal{A}$, define

$$\{A_n \text{ i.o.}\} = \left\{\omega \in \Omega : \sum_n 1_{A_n}(\omega) = \infty\right\} = \bigcap_n \bigcup_{k \geq n} A_k$$

$$\{A_n \text{ ult.}\} = \left\{\omega \in \Omega : \sum_n 1_{A_n^c}(\omega) < \infty\right\} = \bigcup_n \bigcap_{k \geq n} A_k$$

which we call "A_n *infinitely often*" and "A_n *ultimately*".

Definition 13.5: Random Elements and Distributions

Any measurable mapping ξ of Ω into some measurable space $(S, \mathcal{S})$ is called a *random element* in S. If $B \in \mathcal{S}$, then $\{\xi \in B\} = \xi^{-1}B \in \mathcal{A}$, so we can define the push-forward probability measure

$$P\{\xi \in B\} = P(\xi^{-1}B) = (P \circ \xi^{-1})B, \quad B \in \mathcal{S}$$

The probability measure $\mathcal{L}(\xi) = P \circ \xi^{-1}$ on S induced by ξ is called the *distribution* or *law* of ξ.

Definition 14.1: Evaluation Maps

Fixing a measurable space $(S, \mathcal{S})$ and an abstract index set T, write S^T for the class of functions $f : T \to S$, and let $\mathcal{S}^T$ denote the σ-field in S^T generated by all *evaluation maps* $\pi_t : S^T \to S$, $t \in T$, given by

$$\pi_t f = f(t)$$

Definition 14.3: Random Processes

A mapping X as above is called an *S-valued random process on T with paths in $\mathcal{U}$*. By the lemma, we can regard X as a collection of random elements X_t in the *state space* S.

Definition 14.4: Finite-Dimensional Distributions

For any random elements ξ and η in a common measurable space, the equality $\xi \overset{d}{=} \eta$ means that ξ and η have the same distribution, or $\mathcal{L}(\xi) = \mathcal{L}(\eta)$. If X is a random process on some index set T, the associated *finite-dimensional distributions* are given by

$$\mu_{t_1,\dots,t_n} = \mathcal{L}(X_{t_1}, \dots, X_{t_n}), \quad t_1, \dots, t_n \in T,\ n \in \mathbb{N}$$

Definition 14.6: Distribution Functions

For any random vector $\xi = (\xi_1, \dots, \xi_d)$ in $\mathbb{R}^d$, we define the associated *distribution function* F by

$$F(x_1, \dots, x_d) = P\left(\bigcap_{k \le d} \{\xi_k \le x_k\}\right), \quad x_1, \dots, x_d \in \mathbb{R}$$

Definition 14.8: Expected Value

The *expected value*, *expectation*, or *mean* of a random variable ξ is defined as

$$\mathbb{E}[\xi] = \int_\Omega \xi dP = \int_{\mathbb{R}} x(P \circ \xi^{-1})(dx)$$

That is, we can either integrate ξ over Ω, or integrate the possible images with respect to the push-forward measure.

Definition 14.9: Absolute Moments

For any random variable ξ and constant $p > 0$, the integral $\mathbb{E}\,|\xi|^p = \|\xi\|_p^p$ is called the *pth absolute moment* of ξ.

Definition 14.10: Moments

If $\xi \in L^p$ and either $p \in \mathbb{N}$ or $\xi \ge 0$, define the *pth moment* of ξ as $\mathbb{E}[\xi^p]$.

Definition 15.1: Integrable Random Vectors and Processes

A random vector $\xi = (\xi_1, \dots, \xi_d)$ or process $X = (X_t)$ is said to be *integrable* if integrability holds for every ξ_k or X_t. Then, we write

$$\mathbb{E}\xi = (\mathbb{E}\xi_1, \dots, \mathbb{E}\xi_d) \quad \text{or} \quad \mathbb{E}X = (\mathbb{E}X_t)$$

Definition 15.2: Convex Functions

A function $f : \mathbb{R}^d \to \mathbb{R}$ is said to be *convex* if

$$f(px + (1-p)y) \leq pf(x) + (1-p)f(y), \quad x, y \in \mathbb{R}^d,\ p \in [0,1]$$

Definition 15.4: Covariance

The *covariance* of two random variables $\xi, \eta \in L^2$ is given by

$$\operatorname{cov}(\xi, \eta) = \mathbb{E}[(\xi - \mathbb{E}\xi)(\eta - \mathbb{E}\eta)] = \mathbb{E}[\xi\eta] - \mathbb{E}[\xi]\mathbb{E}[\eta]$$

Definition 15.5: Variance

For $\xi \in L^2$, we define the *variance*

$$\operatorname{var}(\xi) = \operatorname{cov}(\xi, \xi) = \mathbb{E}((\xi - \mathbb{E}\xi)^2) = \mathbb{E}\xi^2 - (\mathbb{E}\xi)^2$$

Definition 15.7: Covariance Functions

For any collection of random variables $\xi_t \in L^2$, $t \in T$, the associated *covariance function* $\rho_{s,t} = \operatorname{cov}(\xi_s, \xi_t)$, $s, t \in T$, is *non-negative definite*, in the sense that

$$\sum_{i,j} a_i a_j \rho_{t_i, t_j} \geq 0 \quad \text{for any } n \in \mathbb{N},\ t_1, \ldots, t_n \in T, \text{ and } a_1, \ldots, a_n \in \mathbb{R}$$

Definition 15.8: Mutually Independent Events

Events $A_t \in \mathcal{A}$, $t \in T$ are said to be *mutually independent* if, for any distinct $t_1, \ldots, t_n \in T$,

$$P\left(\bigcap_{k \leq n} A_{t_k}\right) = \prod_{k \leq n} P(A_{t_k})$$

Definition 15.9: Independent Families

More generally, the families $\mathcal{C}_t \subseteq \mathcal{A}$, $t \in T$ are *independent* if independence holds between events A_t for arbitrary $A_t \in \mathcal{C}_t$, $t \in T$.

Definition 15.10: Independent Random Elements

Random elements ξ_t, $t \in T$ are *independent* if independence holds between the σ-fields $\sigma(\xi_t)$, $t \in T$.

Definition 15.13: Separating Classes

Given any set T, we say a class $\mathcal{T} \subseteq 2^T$ is *separating* if, for any $s \neq t$ in T, there exists $S \in \mathcal{T}$ such that exactly one of s and t lies in S.

Definition 16.1: P-Trivial σ-Fields and Degenerate Random Elements

A σ-field $\mathcal{F}$ is said to be *P-trivial* if $PA = 0$ or 1 for every $A \in \mathcal{F}$. A random element is *a.s. degenerate* if its distribution is a degenerate probability measure; that is,

$$P\{\xi \in A\} = \begin{cases} 1, & x \in A \\ 0, & \text{else} \end{cases}$$

for some $x \in S$. In other words, the distribution of ξ is δ_x.

Definition 16.8: Finite Permutations

A *finite permutation* of $\mathbb{N}$ is a bijective map $p : \mathbb{N} \to \mathbb{N}$ such that $p_n = n$ for all but finitely many n.

Definition 16.9: Symmetric Sets

A set $I \subseteq S^\infty$ is said to be *symmetric* (under finite permutations) if

$$T_p^{-1}(I) \equiv \{s \in S^\infty \,:\, s \circ p \in I\} = I$$

for every finite permutation p of $\mathbb{N}$.

Definition 16.10: Permutation Invariant σ-Field

If $(S, \mathcal{S})$ is a measurable space, the symmetric sets $I \in \mathcal{S}^\infty$ form a sub-σ-field $\mathcal{I} \subseteq \mathcal{S}^\infty$, called the *permutation invariant σ-field* in S^∞.

Theorems

Theorem 2.1: Dynkin's π-λ Theorem

If $\mathcal{C}$ is a π-system and $\mathcal{D}$ is a λ-system such that $\mathcal{C} \subseteq \mathcal{D}$, then $\sigma(\mathcal{C}) \subseteq \mathcal{D}$.

Proof Idea: Show that S is a σ-algebra if, and only if, it is both a π and λ system. Now, show $\lambda(\mathcal{C})$ is a σ-algebra by showing it is closed under intersections.
To do this, first show that it is closed under intersections between elements in $\mathcal{C}$ (obvious), then between an element in $\mathcal{C}$ and an element in $\lambda(\mathcal{C})$ by showing it is a λ-system. Then, show that it is closed under intersections by showing that is a λ-system.

Lemma 2.6

If $S_1, S_2, \ldots$ are second-countable topological spaces, then so is $\bigtimes_{t=1}^{\infty} S_t$.

Lemma 2.7

If $S_1, S_2, \ldots$ are second-countable topological spaces, then

$$\mathcal{B}(S_1 \times S_2 \times \cdots) = \mathcal{B}(S_1) \otimes \mathcal{B}(S_2) \otimes \cdots$$

where $\mathcal{B}(X)$ gives the Borel σ-algebra on a topological space X.

Lemma 2.8

Let $f : S \to T$ where $(S, \mathcal{S})$ and $(T, \mathcal{T})$ are measurable spaces. Then,

1. $\mathcal{S}' = f^{-1}(\mathcal{T})$ is a σ-field on S.
2. $\mathcal{T}' = \{B \subseteq T : f^{-1}(B) \in \mathcal{S}\}$ is a σ-field on T.

Proof Idea: Show that preimages behave nicely with set operations.

Lemma 4.3

Considering a mapping f between two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ and let $\mathcal{C} \subseteq \mathcal{P}(T)$ with $\sigma(\mathcal{C}) = T$, then f is $\mathcal{S}\backslash\mathcal{T}$ measurable if, and only if, $f^{-1}(\mathcal{C}) \subseteq \mathcal{S}$

Proof Idea: Let $\mathcal{T}'$ be the sets in T for which the preimage under f is an element of S. We see that $\mathcal{T}' \supseteq \sigma(\mathcal{T}') \supseteq \sigma\mathcal{C}$ as $\mathcal{T}'$ is a σ-algebra.

Lemma 4.4

Let f be a continuous map between two topological spaces S and T with their Borel σ-fields $\mathcal{S}$ and $\mathcal{T}$. Then, f is $\mathcal{S}\backslash\mathcal{T}$ measurable.

Lemma 4.6

Consider any set of functions $f_t : \Omega \to S_t$, $t \in T$ where $(\Omega, \mathcal{A})$ and $(S_t, \mathcal{S}_t)$, $t \in T$ are measure spaces, and define $f = (f_t) : \Omega \to \times_t S_t$. Then, f is $\mathcal{A}\backslash\bigotimes_t \mathcal{S}_t$-measurable if, and only if, f_t is $\mathcal{A}\backslash\mathcal{S}_t$-measurable for all $t \in T$.

Proof Idea: Recall that preimages under products are intersections; leverage this fact and it follows simply.

Lemma 5.2

Let $f_1, f_2, \ldots$, be measurable functions from $(\Omega, \mathcal{A}) \to \overline{\mathbb{R}}$. Then, $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$ and $\liminf_n f_n$ are all measurable.

Proof Idea: Show this holds for $\sup_n f_n$, and then use properties to relate to the other functions.

Lemma 5.3

Let $f_1, f_2, \ldots$, be measurable functions from $(\Omega, \mathcal{A})$ into some metric space (S, ρ). Then,

1. $\{\omega : f_n(\omega) \text{ converges}\} \in \mathcal{A}$ when S is complete.
2. $f_n \to f$ on Ω implies that f is measurable.

Proof Idea: Since S is complete, f_n converges if

$$\lim_{n\to\infty} \sup_{m \geq n} \rho(f_m(\omega), f_n(\omega)) = 0$$

use that ρ is measurable to show that this is a measurable set.

Theorem 5.4: Simple Approximation of Measurable Functions

For any measurable function $f : (\Omega, \mathcal{A}) \to \mathbb{R}_+$, there exists measurable simple functions $f_1, f_2, \ldots : \Omega \to \mathbb{R}_+$ with $0 \leq f_n \uparrow f$.

Proof Idea: Consider

$$f_n(\omega) = 2^{-n} \lfloor 2^n f(\omega) \rfloor \wedge n$$

Lemma 5.5

Fix measurable functions $f, g : (\Omega, \mathcal{A}) \to \mathbb{R}$ and let $a, b \in \mathbb{R}$. Then, $af + bg$ and fg are measurable, and so is f/g if $g \neq 0$ on Ω.

Proof Idea: Write these as limits.

Lemma 6.5: Doob-Dynkin Lemma

Fix two measurable functions f and g on Ω into measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ where the former, $(S, \mathcal{S})$ is Borel. Then, f is g-measurable if, and only if, there exists a measurable $h : T \to S$ such that $f = h \circ g$.

Lemma 6.7

Let μ be a measure on $(\Omega, \mathcal{A})$ and assume that $A_1, A_2, \ldots \in \mathcal{A}$. Then,

1. $A_n \uparrow A$ implies $\mu(A_n) \uparrow \mu(A)$.
2. $A_n \downarrow A$ with $\mu(A_1) < \infty$ implies $\mu(A_n) \downarrow \mu(A)$

Proof Idea: Use differences to write disjointly, and then use complements in A_1 with 1. to prove 2.

Corollary 7.1

Let $\mu_1, \mu_2, \ldots$ be a measure on $(\Omega, \mathcal{A})$ such that $\mu_n \uparrow \mu$ or $\mu_n \downarrow \mu$ with μ_1 bounded. Then, μ is a measure on $(\Omega, \mathcal{A})$.

Lemma 7.2

Let μ and ν be bounded measures on some measurable space $(\Omega, \mathcal{A})$ and let $\mathcal{C}$ be a π-system in Ω such that $\Omega \in \mathcal{C}$ and $\sigma(\mathcal{C}) = \mathcal{A}$. Then $\mu = \nu$ if, and only if, $\nu(A) = \mu(A)$ for all $A \in \mathcal{C}$.

Proof Idea: Show that the set on which they agree is a λ-system.

Lemma 7.4

Let $f \geq 0$ be measurable on a measure space $(\Omega, \mathcal{A}, \mu)$ and let $f_1, f_2, \ldots$ and g be simple measurable functions such that $f_n \uparrow f$ and $0 \leq g \leq f$. Then $\lim_n \mu f_n \geq \mu g$.

Proof Idea: Prove for indicators by considering

$$A_n = \{\omega \in A : f_n(\omega) \geq 1 - \epsilon\}$$

Corollary 7.5

If $f_1, f_2, \ldots, g_1, g_2, \ldots \geq 0$ are measurable simple functions with $f_n \uparrow f$ and $g_n \uparrow f$, then $\lim_n \mu f_n = \lim_n \mu g_n$.

Theorem 7.6: Monotone Convergence Theorem

Let $f, f_1, f_2, \ldots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$ with $0 \leq f_n \uparrow f$. Then, $\mu f_n \uparrow \mu f$.

Proof Idea: Letting $0 \leq g_{n,k} \uparrow f_n$, let

$$h_{n,k} = g_{1,k} \vee \cdots \vee g_{n,k}$$

Lemma 7.7: Fatou's Lemma

For any measurable function $f_1, f_2, \cdots \geq 0$ on $(\Omega, \mathcal{A}, \mu)$, we have

$$\liminf_{n\to\infty} \mu f_n \geq \mu \liminf_{n\to\infty} f_n$$

Proof Idea: Use the fact that $\mu f_m \geq \mu \inf_{k\geq n} f_n$ for $m \geq n$, and use $\inf_{k\geq n} f_k \uparrow \liminf_{k\to\infty} f_k$ for MCT.

Theorem 8.1: Dominated Convergence Theorem

Let $f, f_1, f_2, \ldots$ and $g, g_1, g_2, \ldots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$ with $|f_n| \leq g_n$ for all n and such that $f_n \to f$, $g_n \to g$ and $\mu g_n \to \mu g < \infty$. Then, $\mu f_n \to \mu f$.

Proof Idea: Apply Fatou to $\mu g_n \pm \mu f_n \geq 0$.

Lemma 8.2: Substitution Lemma

Consider a measure space $(\Omega, \mathcal{A}, \mu)$, a measurable space $(S, \mathcal{S})$ and two measurable mappings $f : \Omega \to S$ and $g : S \to \mathbb{R}$. Then,

$$\mu(g \circ f) = (\mu \circ f^{-1})(g)$$

whenever either side exists; that is, when one side exists then so does the other and they are equal.

Thus, the integral of $g \circ f$ with respect to μ is equal to the integral of g with respect to the push-forward measure $\mu \circ f^{-1}$.

Proof Idea: Prove for indicators and then apply linearity and MCT.

Lemma 8.4: Null Sets and Functions

For any measurable function $f \geq 0$ on some measure space $(\Omega, \mathcal{A}, \mu)$, then $\mu f = 0$ iff $f = 0$ μ-a.e.

Lemma 9.1: Completion

Consider a measure space $(\Omega, \mathcal{A}, \mu)$, a σ-field $\mathcal{F} \subseteq \mathcal{A}$, and a Borel space $(S, \mathcal{S})$. Then, a function $f : \Omega \to S$ is $\mathcal{F}^\mu$-measurable if, and only if, there exists some $\mathcal{F}$-measurable function g such that $f = g$ a.e. μ.

Lemma 9.3: Sections

Fix two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a measurable function $f : S \times T \to \mathbb{R}_+$, and a σ-finite measure μ on S. Then,

1. $f(s, t)$ is $\mathcal{S}$ measurable in $s \in S$ for each $t \in T$.
2. $\int f(s, t)\mu(ds)$ is $\mathcal{T}$-measurable in $t \in T$.

Proof Idea: Monotone class argument to show for indicators, then use linearity and MCT.

Theorem 9.4: Lebesgue-Fubini-Tonelli Theorem

For any σ-finite measure spaces $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$, there exists a unique measure $\mu \otimes \nu$ on $(S \times T, \mathcal{S} \otimes \mathcal{T})$ satisfying

$$(\mu \otimes \nu)(B \times C) = \mu(B) \cdot \nu(C), \quad B \in \mathcal{S}, C \in \mathcal{T}$$

Furthermore, for any non-negative measureable function $f : S \times T \to \mathbb{R}_+$,

$$(\mu \otimes \nu)v = \int \mu(ds) \int f(s,t)\nu(dt) = \int \nu(dt) \int f(s,t)\mu(ds)$$

The last relation remains valid for any measurable function $f : S \times T \to \mathbb{R}$ with $(\mu \otimes \nu)\,|f| < \infty$.

Note that these integrals are well defined by the lemma above, and although the inner integrals $\nu f(s, \cdot)$ and $\mu f(\cdot, t)$ may fail to exist on some null sets in S and T, when everything works, everything agrees.

Proof: Uniqueness from monotone class argument and Caratheodory.

Theorem 9.6: Hölder and Minkowski Inequalities

For any measurable functions f and g on some measure space $(\Omega, \mathcal{A}, \mu)$, we have

1. $\|fg\|_r \leq \|f\|_p \|g\|_q$ for all $p, q, r > 0$ with $p^{-1} + q^{-1} = r^{-1}$.
2. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ for all $p \geq 1$, and $\|f + g\|_p^p \leq \|f\|_p^p + \|g\|_p^p$ for $0 < p < 1$.

Proof: Take $r = 1$ and $\|f\|_p = \|f\|_q = 1$. Use Young's inequality

$$ab \leq \int_0^a h(x)dx + \int_0^b h^{-1}(y)dy$$

to the bijective function $y = x^{p-1}$.

For Minkowski, use concavity of x^p.

Lemma 10.1: Completeness

Let (f_n) be a Cauchy sequence in L^p, where $p > 0$. Then $\|f_n - f\|_p \to 0$ for some $f \in L^p$.

Proof: Choose a subsequence $(n_k) \subseteq \mathbb{N}$ with $\sum_k \|f_{n_{k+1}} - f_{n_k}\|_p^{p \wedge 1} < \infty$. By Minkowski's inequality and monotone convergence applied to finite sums, we see that

$$\|\sum_k \left|f_{n_{k+1}} - f_{n_k}\right|\|_p^{p \wedge 1} < \infty$$

so $\sum_k \left|f_{n_{k+1}} - f_{n_k}\right| < \infty$ almost everywhere. Hence, we must have that (f_{n_k}) is Cauchy almost everywhere in $\mathbb{R}$, so by the completeness of $\mathbb{R}$, we see that $f_{n_k} \to f$ for some measurable function f. Then, by Fatou's

lemma, and using the fact that $f = \liminf_{k\to\infty} f_{n_k}$,

$$\begin{aligned}
\|f - f_n\|_p &= \left(\int \left| \liminf_{k\to\infty} f_{n_k} - f_n \right|^p \right)^{1/p} \\
&= \left(\int \liminf_{k\to\infty} |f_{n_k} - f_n|^p \right)^{1/p} \\
&\leq \left(\liminf_{k\to\infty} \int |f_{n_k} - f_n|^p \right)^{1/p} \\
&\leq \liminf_{k\to\infty} \|f_{n_k} - f_n\|_p \\
&\leq \sup_{m\geq n} \|f_m - f_n\|
\end{aligned}$$

which is assumed to go to 0 as $n \to \infty$. Thus, we see that $f_n \to f$ in L^p.

Lemma 10.2: Useful Inequality

If $a, b \geq 0$, then

$$(a+b)^p \leq \begin{cases} a^p + b^p & 0 < p \leq 1 \\ 2^{p-1}(a^p + b^p) & p \geq 1 \end{cases}$$

Lemma 10.3: L^p-convergence

For any $p > 0$, let $f, f_1, f_2, \ldots \in L^p$ with $f_n \to f$ almost everywhere. Then, $f_n \to f$ in L^p iff $\|f_n\|_p \to \|f\|_p$.

Proof: If $f_n \to f$ in L^p, then by Minkowski's

$$\left| \|f_n\|_p^{p\wedge 1} - \|f\|_p^{p\wedge 1} \right| \leq \|f_n - f\|_p^{p\wedge 1} \to 0$$

and so $\|f_n\|_p \to \|f\|_p$. Now, assume the other condition, define

$$g_n = 2^p \left(|f_n|^p + |f|^p \right) \quad \text{and} \quad g = 2^{p+1} |f|^p$$

Then, we see that $g_n \to g$ almost everywhere, and $\mu g_n \to \mu g < \infty$ by our hypotheses (since $g_n < (2f)^p$). Since we see that $|g_n| \geq |f_n - f|^p$, which must go to zero almost everywhere. Thus, $\|f_n - f\|_p^p = \mu |f_n - f|^p \to 0$ by the dominated convergence theorem.

Theorem 10.4: Existence of Orthononal Decomposition

Let M be a closed linear subspace of L^2. Then, any function $f \in L^2$ has any almost everywhere unique decomposition $f = g + h$ with $g \in M$ and $h \perp M$.

Lemma 11.1: Regularity

Let μ be a bounded measure on some metric space S with Borel σ-field $\mathcal{S}$. Then,

$$\mu B = \sup_{F \subseteq B} \mu F = \inf_{G \supseteq B} \mu G, \ B \in \mathcal{S}$$

with F and G restricted to the classes of closed and open subsets of S, respectively.

Lemma 11.2: Approximation

Given a metric space S with Borel σ-field $\mathcal{S}$ and a bounded measure μ on $(S, \mathcal{S})$ and a constant $p > 0$, the set of bounded continuous functions on S is dense in $L^p(S, \mathcal{S}, \mu)$. Thus, for any $f \in L^p$, there exist some bounded continuous functions $f_1, f_2, \ldots : S \to \mathbb{R}$ with $\|f_n - f\|_p \to 0$.

Theorem 12.4: Hahn Decomposition

Any bounded signed measure ν can be written uniquely as a difference of two bounded, non-negative, and mutually singular measures ν_+ and ν_-.

Corollary 12.5: Maximum and Minimum

For any σ-finite measures μ and ν on a common measurable space, there exists a largest measure $\mu \wedge \nu$ bounded by μ and ν (i.e. $\mu \wedge \nu \leq \min\{\mu, \nu\}$) and a smallest measure $\mu \vee \nu$ bounding μ and ν (i.e. $\mu \vee \nu \geq \max\{\mu, \nu\}$). Furthermore,

$$\mu - \mu \wedge \nu \perp \nu - \mu \wedge \nu \quad \text{and} \quad \mu \wedge \nu + \mu \vee \nu = \mu + \nu$$

Theorem 12.7: Radon-Nikodym and Lebesgue Decomposition Theorem

For any σ-finite measures μ and ν on Ω, there exist some unique measures $\nu_a \ll \mu$ and $\nu_s \perp \mu$ such that $\nu = \nu_a + \nu_s$. Furthermore, $\nu_a = f \cdot \mu$ (that is, $\nu_a(A) = \mu(f 1_A)$) for some μ-a.e. unique measurable function $f \geq 0$ on Ω.

Lemma 12.8: Closure

Fix two measures μ and ν on Ω and some measurable functions $f_1, f_2, \ldots \geq 0$ on Ω with $f_n \cdot \mu \leq \nu$. Then $f \cdot \mu \leq \nu$, where $f = \sup_n f_n$.

Lemma 13.1: Partial Density

Let μ and ν be bounded measures on Ω with $\mu \not\perp \nu$. Then, there exists a measurable function $f \geq 0$ on Ω such that $\mu f > 0$ and $f \cdot \mu \leq \nu$.

Note that

$$1\{A_n \text{ i.o.}\} = \limsup_{n \to \infty} 1_{A_n} \quad \text{and} \quad 1\{A_n \text{ ult.}\} = \liminf_{n \to \infty} 1_{A_n}$$

and by Fatou's lemma,

$$P\{A_n \text{ i.o.}\} \geq \limsup_{n\to\infty} PA_n \quad \text{and} \quad P\{A_n \text{ ult.}\} \leq \liminf_{n\to\infty} PA_n$$

Furthermore,

$$P\{A_n \text{ i.o.}\} = \lim_{n\to\infty} P\left(\bigcup_{k\geq n} A_k\right) \leq \lim_{n\to\infty} \sum_{k\geq n} PA_k$$

which gives us the following result.

Lemma 13.4: First Borel-Cantelli Lemma

If $\sum_n PA_n < \infty$, then $P\{A_n \text{ i.o.}\} = 0$.

Lemma 14.2

Fix a measurable space $(S, \mathcal{S})$, an index set T, and a subset $\mathcal{U} \subseteq S^T$. Then, a function $X : \Omega \to \mathcal{U}$ is $\mathcal{U} \cap \mathcal{S}^T$-measurable if, and only if, $X_t : \Omega \to S$ is $\mathcal{S}$-measurable for all $t \in T$.

Proposition 14.5: Finite-Dimensional Distributions

Fix S, T, and $\mathcal{U} \subseteq S^T$, and let X and Y be processes on T with paths in $\mathcal{U}$. Then, $X \overset{d}{=} Y$ if, and only if,

$$(X_{t_1}, \ldots, X_{t_n}) \overset{d}{=} (Y_{t_1}, \ldots, Y_{t_n}), \quad t_1, \ldots, t_n \in T,\ n \in \mathbb{N}$$

Lemma 14.7: Distribution Functions

Let ξ and η be random vectors in $\mathbb{R}^d$ with distribution functions F and G. Then, $\xi \overset{d}{=} \eta$ if, and only if, $F = G$.

For any random element ξ in some measurable space S and an arbitrary measurable function $f : S \to \mathbb{R}$,

$$\begin{aligned}\mathbb{E}[f(\xi)] &= \int_\Omega f(\xi) dP = \int_S f(s)(P \circ \xi^{-1})(ds) \\ &= \int_{\mathbb{R}} x(P \circ (f \circ \xi)^{-1})(dx)\end{aligned}$$

provided one of these three integrals exists. Integrals over a measurable subset $A \subseteq \Omega$ are often denoted by

$$\mathbb{E}[\xi; A] = \mathbb{E}[\xi 1_A] = \int_A \xi dP, \quad A \in \mathcal{A}$$

By Hölder's inequality (or Jensen's, using the convexity of x^p), we see that $\|\xi\|_p \leq \|\xi\|_q$ for $p \leq q$, so the corresponding L^p-spaces are non-increasing in p.

Here is why. We want to show that $(\mathbb{E}\,|\xi|^p)^{1/p} \leq (\mathbb{E}\,|\xi|^q)^{1/q}$. Let $\eta = |\xi|^p$. So, we want to show that $\mathbb{E}[|\eta|] \leq \mathbb{E}[|\eta|^{q/p}]^{p/q}$, or $\mathbb{E}[|\eta|] \leq \mathbb{E}[|\eta|^r]^{1/r}$ where $r = q/p \geq 1$. Then, by Hölder's,

$$\mathbb{E}[|\eta|] = \mathbb{E}[|\eta| \cdot 1] \leq \|\eta\|_r \|1\|_s = \|\eta\|_r = \mathbb{E}[|\eta|^r]^{1/r}$$

where $\|1\|_s = 1$ as we have a probability space, and where we use $p \leq q$ to ensure $r = \frac{q}{p} \geq 1$, so that there exists s with $\frac{1}{r} + \frac{1}{s} = 1$.

Lemma 14.11: Moments and Tails

For any random variable $\xi \geq 0$,

$$\mathbb{E}[\xi^p] = p \int_0^\infty P\{\xi > t\} t^{p-1} dt = p \int_0^\infty P\{\xi \geq t\} t^{p-1} dt, \quad p > 0$$

Proof Idea: Fubini Tonelli swap!

Lemma 15.3: Jensen's Inequality

For any integrable random vector ξ in $\mathbb{R}^d$ and convex $f : \mathbb{R}^d \to \mathbb{R}$, we have

$$\mathbb{E}[f(\xi)] \geq f(\mathbb{E}[\xi])$$

Proof Idea: Tangent line must lie beneath the graph, including the tangent line at $\mathbb{E}[f]$.

Lemma 15.11: Extension

If the π-systems $\mathcal{C}_t$, $t \in T$ are independent, so are the σ-fields $\mathcal{F}_t = \sigma(\mathcal{C}_t)$, $t \in T$.

Corollary 15.12: Grouping

Let $\mathcal{F}_t$, $t \in T$ be independent σ-fields, and let $\mathcal{T}$ be a disjoint partition of T. Then, the σ-fields $\mathcal{F}_S = \bigvee_{t \in S} \mathcal{F}_t$, $S \in \mathcal{T}$ are again independent.

Lemma 15.14: Pairwise Independence

i) The σ-fields $\mathcal{F}_1, \mathcal{F}_2, \ldots$ are independent iff $\bigvee_{k \leq n} \mathcal{F}_k \perp\!\!\!\perp \mathcal{F}_{n+1}$ for all n.

ii) The σ-fields $\mathcal{F}_t$, $t \in T$ are independent iff $\mathcal{F}_S \perp\!\!\!\perp \mathcal{F}_{S^c}$ for all sets S in some separating class $\mathcal{T} \subseteq 2^T$.

Lemma 16.2: Triviality and Degeneracy

A σ-field $\mathcal{F}$ is P-trivial iff $\mathcal{F} \perp\!\!\!\perp \mathcal{F}$. In that case, any $\mathcal{F}$-measurable random element ξ taking values in a separable metric space is a.s. degenerate.

Lemma 16.3: Product Measures

Let $\xi_1, \dots, \xi_n$ be random elements in measurable spaces $S_1, \dots, S_n$ with distributions $\mu_1, \dots, \mu_n$. Then, the ξ_k are independent iff $\xi = (\xi_1, \dots, \xi_n)$ has distribution $\mu_1 \otimes \cdots \otimes \mu_n$.

Lemma 16.4: Conditioning

Let ξ and η be independent random elements in some measurable spaces S and T, and let the function $f : S \times T \to \mathbb{R}$ be measurable with $\mathbb{E}(\mathbb{E}\,|f(s,\eta)|)_{s=\xi} < \infty$. Then,

$$\mathbb{E}f(\xi,\eta) = \mathbb{E}(\mathbb{E}f(s,\eta))_{s=\xi}$$

That is, for each $s \in S$, let $g(s) = \mathbb{E}[f(s,\eta)]$. Then, $\mathbb{E}[f(\xi,\eta)] = \mathbb{E}[g(\xi)]$.

Theorem 16.6: Kolmogorov's 0-1 Law

Let $\mathcal{F}_1, \mathcal{F}_2, \dots$ be independent σ-fields. Then, the tail σ-field $\mathcal{T} = \bigcap_n \bigvee_{k \geq n} \mathcal{F}_k$ is P-trivial.

Corollary 16.7: Sums and Averages

Let $\xi_1, \xi_2, \dots$ be independent random variables, and let $S_n = \xi_1 + \cdots + \xi_n$. Then, each of the sequences (S_n) and (S_n/n) is either a.s. convergent or a.s. divergent. For the latter sequence, the possible limit is a.s. degenerate.

Homework Key Ideas

One line per part: the move that unlocks it.

HW1: Generating Classes

- **1(a)** Each λ-axiom survives intersection, so $\lambda(\mathcal{C})$ is a λ-system. **1(b)** $\lambda(\mathcal{I}) \subseteq \sigma(\mathcal{I})$ always (a σ-field is a λ-system). Dynkin's two-pass argument makes $\lambda(\mathcal{I})$ a π-system, hence a σ-field, so $\sigma(\mathcal{I}) \subseteq \lambda(\mathcal{I})$.
- **2(a)** $\mathbf{1}_\Omega$ is constant (ii); $\mathbf{1}_{A\setminus B} = \mathbf{1}_A - \mathbf{1}_B$ for $B \subseteq A$ (i); $\mathbf{1}_{A_n} \uparrow \mathbf{1}_A$ for increasing unions (iii). **2(b)** $\mathcal{I} \subseteq \mathcal{D}$ by (iv), then π-λ: $\mathcal{F} = \sigma(\mathcal{I}) \subseteq \mathcal{D}$. **2(c)** $f_n = 2^{-n}\lfloor 2^n f \rfloor \wedge n = \sum_{j=1}^{n2^n} 2^{-n}\mathbf{1}_{\{f \geq j2^{-n}\}}$ uses only indicators of $\mathcal{F}$-sets, so $f_n \in \mathcal{H}$ by (b) and (i). **2(d)** By (iii). **2(e)** $f + \|f\|_\infty \geq 0$ is in $\mathcal{H}$ by (d); subtract the constant using (i), (ii).
- **3(a)** $\mathcal{C}' = \mathrm{Span}(\mathcal{C} \cup \{1\}) \subseteq \mathcal{H}$ because $\mathcal{C}$ is closed under products. Zorn: among algebras between $\mathcal{C}'$ and $\mathcal{H}$, ordered by $\subseteq$, a chain's union is an algebra in $\mathcal{H}$, hence an upper bound. **3(b)** The uniform closure of $\mathcal{A}$ is an algebra (bounded functions, so products converge uniformly) and lies in $\mathcal{H}$ by (iv); maximality. **3(c)** Weierstrass: polynomials $p_n \to |x|$ uniformly on $[-\|f\|_\infty, \|f\|_\infty]$; $p_n(f) \in \mathcal{A}$ and $p_n(f) \to |f|$ uniformly; use (b). **3(d)** $f \vee g = \frac{1}{2}(f + g + |f - g|)$, $f \wedge g = \frac{1}{2}(f + g - |f - g|)$. **3(e)** The algebra generated by $\mathcal{A}$ and f is $\{\sum_{k \leq m} a_k f^k : a_k \in \mathcal{A}\}$. Each $a_k^{\pm} f_n^k \in \mathcal{A}$ is ≥ 0 and increases to $a_k^{\pm} f^k$ (with $a_k^{\pm} \in \mathcal{A}$ by (d)), so $a_k f^k \in \mathcal{H}$ by (iii), (i). Hence this algebra lies in $\mathcal{H}$, and maximality gives $f \in \mathcal{A}$. **3(f)** $\mathbf{1}_{A^c} = 1 - \mathbf{1}_A$, $\mathbf{1}_{A\cap B} = \mathbf{1}_A\mathbf{1}_B$; countable unions are increasing limits of finite unions, so use (e). **3(g)** Simple $\mathcal{G}$-functions lie in $\mathcal{A}$; a bounded $f \geq 0$ is an increasing limit of them, so use (e); for general

bounded f, shift by a constant. **3(h)** $g = (f \wedge 1) \vee 0 \in [0,1]$, and $g^n \downarrow \mathbf{1}_{\{f \geq 1\}}$ since $g = 1$ on $\{f \geq 1\}$ and $g < 1$ elsewhere. **3(i)** For $f \in \mathcal{C}$, $c \in \mathbb{R}$: $\{f \geq c\} = \{f - c + 1 \geq 1\}$. With g from (h) for $f - c + 1 \in \mathcal{A}$, $0 \leq 1 - g^n \uparrow \mathbf{1}_{\{f<c\}}$, so (e) gives $\{f \geq c\} \in \mathcal{G}$. These sets generate $\mathcal{F}$. **3(j)** Bounded $\mathcal{F}$-measurable $\subseteq$ bounded $\mathcal{G}$-measurable $\subseteq \mathcal{A} \subseteq \mathcal{H}$, by (i) and (g).

- **4** If $\{C_n\}$ generates Σ, let D be the union of the countable sets among the C_n and C_n^c; D is countable. Each C_n contains or misses $\Omega \setminus D$; such sets form a σ-field, so all of Σ would. But $\{x\}$ with $x \notin D$ does neither.
- **5** Good sets: $\{B \in \sigma(\mathcal{A}) : B \in \sigma(\mathcal{A}') \text{ for some countable } \mathcal{A}' \subseteq \mathcal{A}\}$ contains $\mathcal{A}$ and is a σ-field (a countable union of countable families is countable).

HW2: Atoms, Measurability, Tightness

- **1(a)–(b)** $\omega' \in \text{atom}(\omega) \iff \text{atom}(\omega') = \text{atom}(\omega)$: if some $A_k \ni \omega'$ missed ω, then $A_k^c \ni \omega$ would exclude ω'. So the classes are the atoms, each in $\mathcal{A}$ (countable intersection). **1(c)** Every $A \in \mathcal{A}$ is a union of atoms. Finitely many atoms $\Rightarrow \mathcal{A}$ finite. Infinitely many (countably many, each in $\mathcal{A}$) $\Rightarrow$ the $2^{\aleph_0}$ unions of atoms are distinct sets in $\mathcal{A}$, a contradiction.
- **2** Dyadic cells $I_{n,k} = [k2^{-n}, (k+1)2^{-n})$ (the last one closed). $f(I_{n,k})$ is an interval (IVT), hence Borel. $N_n(y) = \sum_k \mathbf{1}\{y \in f(I_{n,k})\}$ counts cells containing a solution: it is measurable, nondecreasing in n, and $N_n \uparrow N$ as the solutions get separated.
- **3(a)** e_t is continuous since $|f(t) - g(t)| \leq \rho(f,g)$, so $\mathcal{A} \subseteq \mathcal{B}$. **3(b)** By continuity, $\rho(f,g) = \sup_t |e_t(f) - g(t)|$ with t ranging over $\mathbb{Q} \cap [0,1]$, a countable sup. **3(c)** Open balls are in $\mathcal{A}$ by (b); in a separable space every open set is a countable union of balls (or use the hint: closed $F = \{\inf_n \rho(f, g_n) = 0\}$).
- **4(a)** $\bigcup_k B(x_k, 1/n) = S$ by density; continuity from below. **4(b)** $\bigcap_n \bigcup_{k \leq N_n} B(x_k, 1/n)$ is totally bounded, so its closure is too; closed and totally bounded in a complete space $\Rightarrow$ compact. **4(c)** $\mathbb{P}(K^c) \leq \sum_n \epsilon 2^{-n} = \epsilon$.

HW3: Δ, DCT Failure, Completion, Projections

- **1(a)** Check the four cases of $(\mathbf{1}_A, \mathbf{1}_B)$. **1(b)** $|\mathbf{1}_A - \mathbf{1}_C| \leq |\mathbf{1}_A - \mathbf{1}_B| + |\mathbf{1}_B - \mathbf{1}_C|$; integrate.
- **2** Counting measure on $\mathbb{N}$, $f_n = \mathbf{1}_{\{n\}} \to 0$ but $\mu f_n = 1$.
- **3** $B \in \mathcal{T}^\nu \iff B \,\Delta\, B_0 \subseteq N$ with $B_0, N \in \mathcal{T}$, $\nu N = 0$. Then $f^{-1}B \,\Delta\, f^{-1}B_0 \subseteq f^{-1}N$ and $\mu(f^{-1}N) = \nu N = 0$, so $f^{-1}B \in \mathcal{S}^\mu$.
- **4(a)** $h \in N$ and $g - h = (f - h) - (f - g) \perp N$ (since $f - g \perp M \supseteq N$), so h is the projection of g onto N. **4(b)** $\left\langle \widehat{f}, g \right\rangle = \left\langle \widehat{f}, \widehat{g} \right\rangle + \left\langle \widehat{f}, g - \widehat{g} \right\rangle$ and $g - \widehat{g} \perp M \ni \widehat{f}$; same for $\langle f, \widehat{g} \rangle$.

HW4: Convergence and Second Moments

- **1** Fatou on $f_n - g_n \geq 0$ and $h_n - f_n \geq 0$; $|f| \leq |g| + |h|$ gives integrability.
- **2** $\text{var}\,|f_n| \leq \text{var}\, f_n \leq M\mu|f_n|$ (since $|\mu f_n| \leq \mu|f_n|$). Let $a_n = \mu|f_n|$. *Case* $\limsup a_n < \infty$*:* with $C = \sup_n a_n$, Chebyshev gives $\mu(|f_n| > K) \leq MC/(K - C)^2$, so $\mu\{|f_n| > K \text{ ult.}\} \leq MC/(K-C)^2 \to 0$ as $K \to \infty$; hence $\liminf |f_n| < \infty$ a.e. *Case* $a_{n_k} \to \infty$*:* $\mu(|f_{n_k}| \leq a_{n_k}/2) \leq 4M/a_{n_k} \to 0$, so $\mu\{|f_{n_k}| > a_{n_k}/2 \text{ i.o.}\} \geq \limsup = 1$; hence $\limsup |f_n| = \infty$ a.e.
- **3** With $A = \{|f| \geq \lambda \|f\|_1\}$: $\|f\|_1 = \mu(|f|; A^c) + \mu(|f|; A) \leq \lambda\|f\|_1 + \|f\|_p\, \mu(A)^{1/q}$ (Hölder on A); rearrange.
- **4** Problem 3 with $p = q = 2$: $\mu\{|f_n| \geq \lambda\|f_n\|_1\} \geq (1-\lambda)^2\beta^{-2}$, and $\lambda\|f_n\|_1 \geq \lambda\alpha\beta^{-1}$. Finish with $\mu\{A_n \text{ i.o.}\} \geq \limsup \mu A_n$.

HW5: Total Variation, Hellinger, Binary Expansions

- **1(a)** $\lambda = \mu + \nu$ and Radon–Nikodym; the densities can be taken in $[0,1]$. **1(b)** Compare any λ with $\kappa = \mu+\nu$: $\kappa = h\cdot\lambda$ with $h = f+g$, and $f = f_\kappa h$, $g = g_\kappa h$ λ-a.e. Both integrands $\phi(f,g)$ are 1-homogeneous, so $\lambda\,\phi(f,g) = \lambda\big(h\,\phi(f_\kappa, g_\kappa)\big) = \kappa\,\phi(f_\kappa, g_\kappa)$. **1(c)** They are the $L^1(\lambda)$ distance of f, g and the $L^2(\lambda)$ distance of $\sqrt{f}, \sqrt{g}$; use one λ dominating all three measures for the triangle inequality. **1(d)** $(\mu-\nu)_\pm = (f-g)^\pm\cdot\lambda$; the sup is attained at $A = \{f \ge g\}$. **1(e)** $(\sqrt{f}-\sqrt{g})^2 \le |\sqrt{f}-\sqrt{g}|(\sqrt{f}+\sqrt{g}) = |f-g|$. Other side: Cauchy–Schwarz on $|f-g| = |\sqrt{f}-\sqrt{g}|(\sqrt{f}+\sqrt{g})$, then Minkowski: $\|\sqrt{f}+\sqrt{g}\|_2 \le (\mu\Omega)^{1/2} + (\nu\Omega)^{1/2}$.

- **2(a)** $|\mu_n h - \mu h| \le \|h\|_\infty\, d_{\mathrm{TV}}(\mu_n, \mu)$. **2(b)** $\nu A = 0 \Rightarrow \mu_n A = 0 \Rightarrow \mu A \le d_{\mathrm{TV}}(\mu_n,\mu) \to 0$. **2(c)** $\lambda = \sum_n 2^{-n}\mu_n/(1+\mu_n\Omega)$ dominates every μ_n; the densities are Cauchy in $L^1(\lambda)$, which is complete; the limit $f \ge 0$ gives $\mu = f\cdot\lambda$.

- **3(a)** The preimage of a length-k cylinder is a dyadic interval $[m2^{-k}, (m+1)2^{-k})$; cylinders generate Σ, so Lemma 4.3 applies. **3(b)** 2^{-k}.

- **4(a)** $G_f = \varphi^{-1}\{0\}$ with $\varphi(x,y) = y - f(x)$, measurable by Lemmas 4.6, 5.5 and $\mathcal{B}(\mathbb{R}^{d+1}) = \mathcal{B}(\mathbb{R}^d)\otimes\mathcal{B}(\mathbb{R})$. **4(b)** 0, by Tonelli: $\lambda^{d+1}(G_f) = \int \lambda^1(\{f(x)\})\,dx = 0$.

HW6: Borel–Cantelli Variants

- **1(a)** $\bigcup_{k\ge n} A_k \setminus \bigcup_{k\ge n}(A_k \cap A_{k+1}^c) \subseteq \{A_k \text{ ult.}\}$, and $\mathbb{P}\{A_k \text{ ult.}\} \le \liminf \mathbb{P}A_k = 0$. So $\mathbb{P}\{A_n \text{ i.o.}\} \le \sum_{k\ge n}\mathbb{P}(A_k \cap A_{k+1}^c) \to 0$. **1(b)** If $\mathbb{P}\{A_n \text{ i.o.}\} < 1$, some $A = \bigcap_{k\ge N} A_k^c$ has $\mathbb{P}A > 0$; but $A_n \cap A = \emptyset$ for $n \ge N$, so $\sum_n \mathbb{P}(A_n \cap A) < \infty$.

- **2(a)** Borel–Cantelli I on $A_n \cap A$: $\mathbb{P}(\{A_n \text{ i.o.}\} \cap A) = 0$. **2(b)** Pick N with $\mathbb{P}(\bigcup_{k=n}^N B_k) > 1-\epsilon$, and let K be the last $k \in [n, N]$ with B_k. Then $\{K = k\} = B_k \cap B_{k+1}^c \cap \cdots \cap B_N^c \in \sigma(B_k, B_{k+1}, \dots)$ is independent of A_k, so $\mathbb{P}\big(\bigcup_{k=n}^N A_k \cap B_k\big) \ge \sum_k \mathbb{P}A_k\,\mathbb{P}(K=k) \ge (\inf_{k\ge n}\mathbb{P}A_k)(1-\epsilon)$. Let $\epsilon \to 0$, then $n \to \infty$.

- **3(a)** Identify ϑ with its set S of $+1$'s. $S \subsetneq S'$ raises the sum by $2\sum_{S'\setminus S} x_k \ge 2 > h$, so the ϑ's landing in $[a, a+h]$ form an antichain; Sperner, and each ϑ has probability 2^{-n}. **3(b)** Symmetric $\xi \overset{d}{=} \vartheta|\xi|$ with $\vartheta = \pm 1$ fair and independent of $|\xi|$. Condition on $(|\xi_k|)$ (Lemma 16.4). Since $|\xi_k| > c$, rescale by c and cover $[a, a+h]$ by $\lfloor h/c\rfloor + 1$ intervals of length $c < 2c$: the bound is $(\lfloor h/c\rfloor+1)2^{-n}\binom{n}{\lfloor n/2\rfloor} \to 0$, uniformly in a. Take $[a, a+h] = [-r, r]$ for the last claim.

- **4(a)** Cauchy–Schwarz on $N = \sum_k \mathbf{1}_{A_k}$: $\mathbb{E}N = \mathbb{E}[N; N>0] \le (\mathbb{E}N^2)^{1/2}\,\mathbb{P}(N>0)^{1/2}$, with $\mathbb{E}N = \sum_k \mathbb{P}A_k$ and $\mathbb{E}N^2 = \sum_k\sum_\ell \mathbb{P}(A_k \cap A_\ell)$. **4(b)** For $A_m, \dots, A_n$ with $s = \sum_{k=m}^n \mathbb{P}A_k$: $\sum_k\sum_\ell \mathbb{P}(A_k \cap A_\ell) \le s + s^2$, so $\mathbb{P}\big(\bigcup_{k=m}^n A_k\big) \ge s/(1+s) \to 1$ as $n \to \infty$, for every m.

Standard Computations and Facts

These are the reusable calculations behind the 2025 exam and HW5–6.

Coin Tossing and Random Walks

- **Fair coin measure on $\{0,1\}^{\mathbb{N}}$:** $\mathbb{P}(\text{first } n \text{ coordinates} = x_1, \dots, x_n) = 2^{-n}$; it is determined by cylinders (a π-system containing Ω). The binary-expansion map pushes Lebesgue measure on $[0,1)$ to it (HW5 #3), so the binary digits of a uniform point are i.i.d. fair bits.
- **Biased coin P_p on $\{\pm 1\}^{\mathbb{N}}$:** one path $x_1, \dots, x_n$ with sum s has probability $(pq)^{n/2}(p/q)^{s/2}$; its likelihood ratio to the fair coin is $(4pq)^{n/2}(p/q)^{s/2}$.
- **Exit from $(-a,a)$:** $T < \infty$ a.s.; $P_p\{S_T = a\} = p^a/(p^a + q^a)$; T and S_T are independent; the law of T under P_p is the tilt of its law under $P_{1/2}$ by $z = 2\sqrt{pq}$, and it decreases stochastically as p moves away from $1/2$.
- **Blocks:** disjoint blocks of i.i.d. coordinates are independent (grouping, Lemma 15.12); a fixed pattern of length M with probability $p^M > 0$ occurs in some block a.s., since $\prod_k (1 - p^M) = 0$.
- **Symmetry:** $\omega \mapsto -\omega$ preserves the fair-coin measure; check on cylinders, extend by π-λ.

Orders and Tilts

- **Tilt** $\pi_z\{i\} \propto z^i \pi\{i\}$: $z' \le z'' \Rightarrow \pi_{z'} \preceq \pi_{z''}$ (Harris applied to f and $h(i) = (z''/z')^i$, truncated).
- **Stochastic order:** $\mu \preceq \nu \Rightarrow \mu[x,\infty) \le \nu[x,\infty)$; take $f = \mathbf{1}_{[x,\infty)}$.

Independence Checks

- **Discrete variables:** independent $\iff \mathbb{P}\{X = x, Y = y\} = \mathbb{P}\{X = x\}\,\mathbb{P}\{Y = y\}$ for all x, y; sum over $x \in A$, $y \in B$ by countable additivity.
- **Random vectors:** independent $\iff$ the joint distribution function factorizes (orthants form a π-system, and both sides are probability measures).
- **Events:** A, B independent $\Rightarrow A, B^c$ independent (and A^c, B^c), since $\{A\}$ and $\{B\}$ are π-systems and Lemma 15.11 extends independence to $\sigma(A) = \{\emptyset, A, A^c, \Omega\}$ and $\sigma(B)$.

Null Sets and Measurability Facts

- Countable sets, graphs of measurable functions (HW5 #4) and the Cantor set are Lebesgue-null.
- **Vitali:** there is no rotation-invariant probability on all subsets of the circle S^1 (circumference 2π). Pick one point A per family $\omega \sim \omega + n$, $n \in \mathbb{Z}$; the disjoint translates $A + n$ cover S^1, forcing $1 = \sum_{n \in \mathbb{Z}} \mathbb{P}(A)$, which is impossible.
- Atoms of a countably generated σ-field are measurable; the countable-cocountable σ-field on $[0,1]$ is not countably generated (HW1 #4).
- On $C[0,1]$ with the sup metric, the Borel σ-field equals $\sigma(\text{evaluations})$ (HW2 #3). Probability measures on Polish spaces are tight (HW2 #4).
- **Tail events:** $\{S_n \text{ converges}\}$, $\limsup S_n/n$, $\{\xi_n \in B \text{ i.o.}\}$. **Symmetric but not tail:** $\{S_n = 0 \text{ i.o.}\}$, $\{S_n > 0 \text{ i.o.}\}$.

2025 Midterm: Solution Sketches

Setup. $\Omega = \{-1,+1\}^{\mathbb{N}}$, $P_p = \mu_p^{\otimes\mathbb{N}}$ with $\mu_p\{+1\} = p$, $q = 1-p$, $\xi_k(\omega) = \omega_k$, $S_n = \sum_{k\le n}\xi_k$, $T = \inf\{n \in \mathbb{N} : S_n \in \{-a,+a\}\}$. π is a probability on $\mathbb{N} = \{1,2,\dots\}$ and $\pi_z\{i\} = z^i\pi\{i\}/Z(z)$ with $Z(z) = \sum_j z^j\pi\{j\}$, $z \in (0,1]$. $\mu \preceq \nu$ means $\mu f \le \nu f$ for all bounded, Borel, nondecreasing f.
Answers to remember: $P_p\{S_T = a\} = p^a/(p^a+q^a)$, and the law of T under P_p is the tilt of its fair-coin law by $z = 2\sqrt{pq}$.

- **(1) Harris inequality.** For all x', x'' the factors $f(x') - f(x'')$ and $g(x') - g(x'')$ have the same sign, so the integrand is ≥ 0. It is bounded and $\mu\otimes\mu$ is a probability, so Fubini lets you expand:

$$0 \le \iint \big(f(x') - f(x'')\big)\big(g(x') - g(x'')\big)\,\mu\otimes\mu(dx', dx'') = 2\mu(fg) - 2\,\mu f\,\mu g.$$

- **(2) Tilting is monotone.** Let $r = z''/z' \ge 1$ and $h(i) = r^i$, which is nondecreasing. Then $\pi_{z''}\{i\} = h(i)\pi_{z'}\{i\}/\pi_{z'}h$, so $\pi_{z''}f = \pi_{z'}(fh)/\pi_{z'}h$. Extend f to $\mathbb{R}$ as a nondecreasing step function. Since h may be unbounded, apply (1) to $h_N = h\wedge N$: $\pi_{z'}(fh_N) \ge \pi_{z'}f\cdot\pi_{z'}h_N$. Let $N\to\infty$ by DCT ($|fh_N| \le \|f\|_\infty h$ and $\pi_{z'}h = Z(z'')/Z(z') < \infty$) to get $\pi_{z''}f \ge \pi_{z'}f$.

- **(3) Likelihood ratio.** If $s = \sum_k x_k$, the path has $(n+s)/2$ plus signs and $(n-s)/2$ minus signs, so its P_p-probability is $p^{(n+s)/2}q^{(n-s)/2} = (pq)^{n/2}(p/q)^{s/2}$. Divide by $P_{1/2} = 2^{-n}$.

- **(4) $T < \infty$ a.s.** Take $M = 2a$ and $A_k = \{\text{all } M \text{ steps in block } k \text{ are } +1\}$. The blocks are disjoint, so the A_k are independent (grouping) with $P_p(A_k) = p^M > 0$, and $P_p(\text{no } A_k \text{ with } k \ge m) = \lim_K (1-p^M)^K = 0$. On A_k: if $T > (k-1)M$ then $|S_{(k-1)M}| < a$, the block adds $2a$, and ±1 steps cannot jump over $+a$, so $T \le kM$.

- **(5) Symmetry.** $\varphi(\omega) = -\omega$ preserves $P_{1/2}$ (every length-n cylinder has probability 2^{-n}; extend by π-λ). $T\circ\varphi = T$ and $S_T\circ\varphi = -S_T$, so $\{T = n, S_T = a\}$ and $\{T = n, S_T = -a\}$ have equal probability and partition $\{T = n\}$.

- **(6) Independence.** $\{T = n, S_T = a\}$ is a union of length-n paths with $\sum x_k = a$. Sum (3) over them and use (5):

$$P_p\{T = n, S_T = \pm a\} = \tfrac{1}{2}(4pq)^{n/2}(p/q)^{\pm a/2}P_{1/2}\{T = n\}.$$

  Write $\rho = (p/q)^{a/2}$, $c_n = \frac{1}{2}(4pq)^{n/2}P_{1/2}\{T = n\}$, $C = \sum_n c_n$. Then $P_p\{T = n\} = c_n(\rho + \rho^{-1})$, $P_p\{S_T = \pm a\} = C\rho^{\pm1}$, and (4) forces $C(\rho+\rho^{-1}) = 1$. So $P_p\{T = n\}\,P_p\{S_T = a\} = c_n\rho = P_p\{T = n, S_T = a\}$, and likewise for $-a$. *State the criterion:* for discrete X, Y, a factorizing joint pmf gives $\mathbb{P}\{X\in A, Y\in B\} = \mathbb{P}\{X\in A\}\,\mathbb{P}\{Y\in B\}$ by summing over the countably many points. Also $P_p\{S_T = a\} = \rho^2/(\rho^2+1) = p^a/(p^a+q^a)$.

- **(7) Stochastic order of T.** By (6), $P_p\{T = n\} \propto z^n P_{1/2}\{T = n\}$ with $z = 2\sqrt{pq} \in (0,1]$, so the law of T under P_p is π_z with π the law of T under $P_{1/2}$ (a probability on $\mathbb{N}$ by (4)). pq decreases on $[\frac{1}{2},1)$, so $\frac{1}{2} \le p'' < p' < 1$ gives $z' < z''$, and (2) gives $\pi' \preceq \pi''$: the more biased walk exits sooner.